Junseok Lee

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EDUCATION

University of Toronto — **Incoming, September 2026**
Master of Engineering in Electrical & Computer Engineering — *Toronto, Ontario, Canada*
- Focus: **Robotics, Control Systems, Embedded Systems, and Intelligent Automation**

McMaster University — **Completed, April 2026**
Bachelor of Engineering in Electrical & Computer Engineering (Co-op) — *Hamilton, Ontario, Canada*
- Graduated with **Cumulative GPA: 3.2/4.0**
- **Relevant Coursework**: Logic Design, Digital Signal Processing, Circuits and Waves, Programming, Hardware Design, Web and System Development, Control Systems, SCADA Systems

SKILLS

- Programming Languages: C, C++, Python, Java, JavaScript, Verilog (FPGA), Bash, HTML/CSS
- Control & Embedded Systems: CODESYS (PLC), CAN Protocol, SCADA Systems, Arduino, Danfoss Plus+1 Guide & Service Tool, Embedded Systems, Hardware Debugging, PLC Programming, HMI Development
- Engineering Tools: AutoCAD, SolidWorks, PSpice, LTspice, Electric Circuit Design

WORK EXPERIENCE

Flodraulic | *Georgetown, Canada* | *CODESYS, CAN Protocol, Danfoss Plus+1, AutoCAD, Flutter* — **Aug 2024 - Aug 2025**
- Designed and implemented HMI system, **integrating control system and hydraulics** using CODESYS, AutoCAD, Danfoss +1 Guide/Service tool (Logic Design tool) as a Control System Engineer**.**
- Diagnosed and resolved **wireless system issues** (transmitters, receivers, internal logic, cable harnesses), **reducing troubleshooting time by 40%**.
- Conducted **hardware diagnostics** and implemented customer-requested program **logic updates onsite** to meet specific functional requirements.
- Developed **custom debugging solutions** tailored to client needs using **Flutter**, improving diagnostic accuracy and **enhancing user experience** for the company's app.
- Managed **data visualization** with **Adobe XD, Illustrator, and Microsoft Excel**, streamlining reporting processes.

SROOK Pay | *South Korea* | *Google Analytics, Java, JavaScript, MySQL, HTML/CSS* — **June 2024 - Aug 2024**
- Developed **integrated website reports** using **Java and JavaScript**, optimizing system performance and design.
- Analyzed consumer behavior by managing and retrieving **Google Analytics** data through **MySQL** and **Google Cloud**, improving data-driven decision-making.

TECHNICAL PROJECTS

Fitting-room Intelligent Tracking & Transport | *Flutter, Python, STM32, FreeRTOS, AI/CV, MQTT* — **Sep 2025 – Apr 2026**
- Developed a **retail automation system** with an autonomous rover for fitting-room assistance and clothing delivery.
- Built customer and staff interfaces for queue tracking, room status monitoring, and service requests.
- Integrated STM32, FreeRTOS, and sensors to support safe rover operation in retail environments.

Integrated Automation Control System | *Python, Arduino, CAN BUS interface* — **May 2025 – July 2025**
- Developed an **Arduino-based control system** at Flodraulic, optimizing **real-time data handling and device operability via CAN bus** with dynamic protocol and baud rate adjustments.
- Engineered a real-time CAN bus system with dynamic protocol selection, improving communication efficiency.
- Achieved an ***80%* reduction** in device testing time through embedded diagnostics and enhanced LED/button feedback.

Pacemaker | *Pacemaker, Python, MATLAB Simulink* *GitHub* — **Sep 2023 – Dec 2023**
- Designed a **Digital Circuit Model (DCM)** to simulate the bioelectrical interface between a heartbeat programming application (built with **Python Tkinter**) and a pulse generator.
- Programmed the pacemaker's functionality for multiple operational modes and integrated **real-time electrogram simulations** using **MATLAB Simulink**.

Hardware Implementation of an Image Decompressor | *Verilog, C* — **Sep 2023 – Dec 2023**
- Developed **custom decoding circuitry** to process. **mic17 compressed images**, utilizing UART for data transfer and SRAM for storage and applied **Color Space Conversion and Inverse Discrete Cosine Transform (IDCT)** to enhance image quality while optimizing **memory and processing efficiency**.

LEADERSHIP/ACTIVITIES

Korean Students Association of Canada (KSAC) | Head of IT — **June 2022 – Present**
- Managed website maintenance, security updates, and feature development to improve platform stability and user experience. (https://ksaccanada.com/)